Spring 2024 – User Research Methods – Professor Charles Raffaele
Think Aloud Protocol Design Document REFLECTION – Darryl Hutchinson

INTRODUCTION & RESEARCH QUESTIONS

Though they have great potential for entertainment and learning, video games are typically very complex and it can be easy for inexperienced players to get overwhelmed and lose site of the objectives the designer wants them to experience. To better understand intuitive level design and which game elements do the best job of capturing player attention regardless of experience level such that they don't get lost and can progress through an experience, I want to analyze the upcoming video game, Europa.

Designed and developed by Helder Pinto and Novadust studios, Europa is a bright and colorful Sci-Fi adventure game with a heavy influence of painterly Ghibli style visual animation, smooth freeform movement and traversal with low stress exploration. The game has been in development since 2017 and is currently available in demo form with a scheduled release window of spring 2024.

I am interested in exploring pacing/pathing and environmental storytelling. I would like to know specifically how game elements in the game world can communicate which ways players can travel without using text or unnatural UI markers. Since video games are relatively complex bundles of systems that are easy to get lost within, I want to try and understand how to better lead the player along an experience with natural elements that help communicate objectives while maintaining immersion~~help maintain immersion and still communicate.~~

The target audience for this research could be Novadust studios themselves to gain more perspective on what their players are experiencing. With more feedback about what is drawing player attention, adjustments can be made to the level design to further avoid players getting lost or confused in the game. Another audience could also be other developers, designers, critics and/or researchers who want to more deeply understand how specific elements in Europa create specific responses or perceptions in players. ~~Using t~~This analysis as a case-study towards some generalizable level design techniques have also been revealing ~~may reveal applications~~ insights that may prove valuable to ~~to~~ other future projects for me and/or other game designers.

The results of this analysis will be submitted to Professor Raffaele for evaluation.

Research Questions:

1. **How does an inexperienced player recognize and follow the critical path through a game world**? So we can better understand if the critical path is clear and understood.
    a. **Think-Aloud**: What utterances reflect player thought as they identify and follow/deviate from the critical path?
2. **How does an inexperienced player spend their time in the game world?** In order to understand what the player finds interesting.
    a. **Think-Aloud**: What utterances reflect player thoughts on the way to/from points of interest?
    b. **Think-Aloud**: What utterances reflect player thoughts while at a point of interest?

3. **How does the player manage challenge/enjoyment throughout the experience?** So we can better understand what is getting most of the player's attention.
    a. **Think-Aloud**: At specific points of interest where more time is spent, what utterances reflect the respondent thinking about whatever they are focused on?

STUDY DESIGN

The think aloud protocol is a useful approach for getting unfiltered and immediate responses from participants as they experience some treatment. Unlike observation that emphasizes what participants do and the actions they take, think-aloud protocols allow us to get a closer look at what respondents are thinking as they make decisions in response to various stimuli. I would like to clearly express here that I have seen no evidence pointing to the idea that thought in general, let alone human thought, is thoroughly understood. Many researchers in this field make the claim that think-aloud is an accurate proxy for thought and therefore a valid tool for user researcher. I currently cannot and do not make a similar jump in reasoning regarding how closely think-aloud output resembles actual thought. Even without that however, I can still accept that think-aloud protocols have great value in user researcher in general.

Limitations aside, I will employ the use of think-aloud as the best tool available for me to get closer to real-time participant thought processes. With specific regard to the game being tested, Europa, think-aloud will be used to expose deeper insights into what the player is thinking while they identify points of interest, traverse from point to point, and make decisions about where to go and how they navigate challenges and enjoyment across the virtual landscape. What do they express when they see an open field with ruins placed conveniently in the center of the frame? What are they expressing when they look passed a landmark and instead wander off in another direction? What are they expressing when they struggle with a particular action or miss a prompt? Think-aloud isn't a perfect capture of pure thought, but the utterances that come as a result of thought are still valuable reflections of the player's perspective.

Before the participant start the Europa demo, I will also conduct a brief think-aloud warm up session. Then I will start the demo, allow the participant to play and try to take notes on what they say via a think-aloud form (**see figure 1 below**). Then I will thank them for their time and let them leave. I do not plan on conducting a retrospective think-aloud analysis, but I do intend to record the entire session from warm-up trough to the main play-test.

Utterances to be captured will include:

- RQ1/RQ2/RQ3 – Thoughts uttered around points of interest:
    - While approaching
    - Upon arrival
    - While departing
- RQ1/RQ2/RQ3 – What do players think/express when they dwell at a point of interest.

*Additional variables to be measured if provided access to debug tools:

- RQ1/RQ2 - How do utterances plot onto vector map of the path player takes through the game

- RQ1/RQ2 - Number/nature of utterances when player deviates from the critical path
- RQ3 – Do utterances change as players deviate further from the critical path

Instruments will include:

- Learner profile gaming experience pre-questionnaire
- Think Aloud - audio recording device
    - Think-aloud recording data
    - Think-aloud quick response form
- Semi-structured gaming experience post-interview
- *User Position Logs* (If provided debug access)

PARTICIPANT PERSONA

This participant is a personal relationship. They have little to no experience with video game culture, is 70 years old, identifies as male and his highest level of formal education is college. The participant is also now retired. The participant claims not to play video games on any regular basis and has little to no familiarity with modern game design.

Sample Think Aloud Form

	A	B	C	D	E	F	G	H	I	J
1	**SAMPLE FORM**	Grave Cave	To next POI	Ruins	To next POI	Tree	To next POI	Broken Bridge	To next POI	Pond
2	# of utterances									
3	eg + positive									
4	- negative									
5	/ neutral									
6										
7										
8										
9	Utterance topic									
10										
11										
12										
13										
14										
15	**Action Codes**	J	St	Wa	Lfl	sk	en	un	cl	
16		Jump	Stuck	Wander	Looking For interaction	Skipped	Enjoy	unclear	clear	
17	E.G.	j on turtle	st @first gap	wa @ windmill	lfi @ doors	sk POI3	en colors	prompt #1 un	bridge area cl	

	A	B	C	D	E	F	G	H	I	J
1	**SAMPLE FORM**	POI#1_____	To next POI	POI#2_____	To next POI	POI#3___	To next POI	POI#4_____	To next POI	POI#5_____
2	# of utterances									
3	eg + positive									
4	- negative									
5	/ neutral									
6										
7										
8										
9	Utterance topic									
10										
11										
12										
13										
14										
15	**Action Codes**	J	St	Wa	Lfl	sk	en	un	cl	
16		Jump	Stuck	Wander	Looking For interaction	Skipped	Enjoy	unclear	clear	
17	E.G.	j on turtle	st @first gap	wa @ windmill	lfi @ doors	sk POI3	en colors	prompt #1 un	bridge area cl	

Figure 1

REFLECTIONS

Problems with Think-Aloud form

1. Printer decided to stop working so right before the session, so I ended having to rely on hand written memos
2. Think-Aloud Form structure I ended up with inadvertently assumes the player will traverse the game space in a relatively linear fashion, but the participant was actually wandering and back-tracking significantly more than I anticipated. The current think-aloud form design makes a wandering player's actions more difficult to quickly capture.

Potential Solutions

1. Start preparing documents further ahead of time and also have at least two hand written copy read in case of a emergency (one to use for the current session and one to leave blank to photo copy later)
2. I can redesign the form without specific landmarks that I assume the player will encounter and perhaps instead I can leave black spaces to fill in according to how each player proceeds. Leaving the form as POI#1, POI#2, etc. will require more writing during the session, but It should be easier to capture potentially varying paths through the game relative to each participants' unique path

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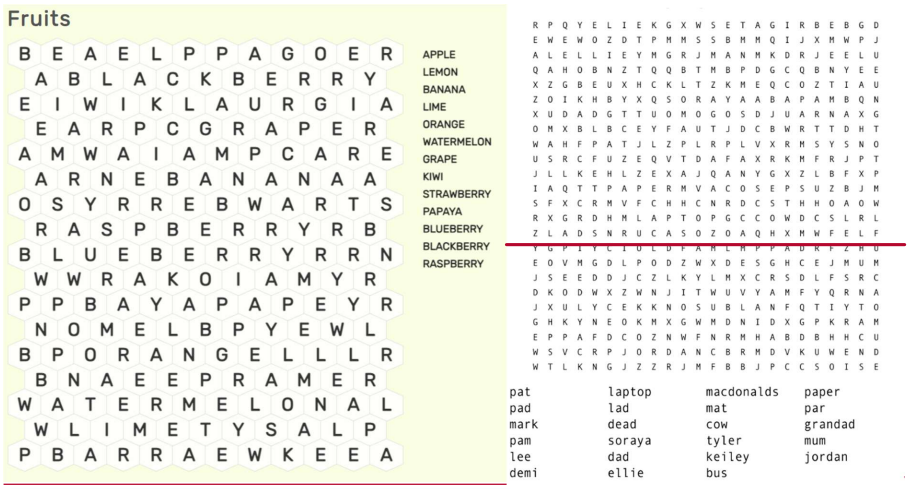
THINK-ALOUD PROCEDURE

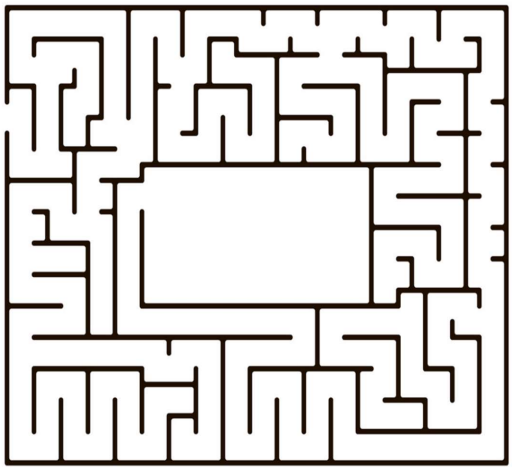
Warm up

When the participant arrives, they will be greeted, asked to sit down and then the warm up can begin. The warm-up section will include two parts.

Warm up part 1: A brief 2min period of casual questions to help elicit comfort and honest expression. Potential topics may include but are not limited to: the weather, current events, hobbies, work, etc. Discourse about video games will be avoided so as not to inadvertently influence the play-test. Thereafter, warm up # 2 can begin.

Warm up part 2: A quick briefing of how researchers would like the participants to engage with the think-aloud protocol will commence. Think aloud expectations will be explained and if the participants needs more support the think-aloud protocol can be demonstrated and/or time provided for them to practice with a simple line maze or crossword puzzle. Complete the puzzle while employing think-aloud protocols.





Figures 2 & 3, a simple word search puzzle and/or a simple line maze puzzle can be used to warm-up with think-aloud protocol.

REFLECTIONS

The warm up I think went well, I decided to only use a different word search puzzle, but I think I was still able to describe how I wanted the participant to think-aloud and provide them space to practice with this activity prior to the main session. However, one potential issue arose.

Problem with warm-up activity:

1. One thing I noticed, was that the participant was doing a lot of internal thinking. And even after being prompoted to continue speaking, the participants would stop mid-sentence to focus on the task. The task may have been too easy to the point that progressing didn't require many

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steps that were easy to verbalize and instead was just a simple task of searching. I tried to prompt them to verbalize specifically what they were looking for next, what letters they were focused, what their strategy was,etc. It was okay overall, but there were still many gaps in their verbalization.

Potential Solution

1. Perhaps a slightly more difficult task would be more suitable for a warm-up. Not so difficult that the participant gets stressed or tired, but something that requires more steps to solve might provide the participant more opportunities to express though processes. The other puzzle I originally proposed might fall into the same category of a being a bit too easy, so I can reseach other activities. Some possibilities might include: simlpe arithematic questions, that might require a few steps to complete, a simple card game like go fish or "I Declare War", or simple where's waldo picture search might also work well. There is much room for testing different approaches here.

Set-up Instructions

1. Prepare the test setup (desk, chair, PC with mouse and keyboard)
2. Prepare the recording setup (audio recording device, think-aloud response form)
3. Prepare warm up questions and/or warm up worksheets if necessary
4. Participant will be reminded to try and say anything that comes to mind as they during the session (including warm-up, main test, questionnaires)
    a. Think-aloud practice with puzzle worksheets if necessary.
5. Once everything has been confirmed to be in working order, commence the session and start recording

REFLECTION

Besides the problem mentioned earlier regarding, printer issues, none of the mistakes from the first observation session were not repeated and the setup was better prepared for the participants, however I noticed a minor issue.

Problem:

1. The controller was placed on the table in between the participant and the computer and after the participant sat down, I started to describe the warm-up activity, but the participant immediately reached for the controller before it was necessary.

Potential Solution:

1. Next time, I can keep the monitor closed/off and/or move the controller away from the player during the warm-up activity so the first task is a little more clear and participant attention and cognitive load are both better managed.

General Probes

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Because thinking aloud while playing a game isn't necessarily standard behavior for everyone, and the participant may forget to keep talking, some general probes will be provided here for quick reference:

- What is happening in this area?
- What do you think about the presentation in this area?
- What is your plan in this area?
- What stands out to you in this section?
- What do you see now?
- What did you just do?
- Why did you do that?
- What are you trying to do next?

Task-specific probes

Because thinking aloud while playing a game isn't necessarily standard behavior for everyone, and the participant may forget to keep talking, some task-specific probes will be provided here for quick reference:

- What instructions were in the last prompt?
- Have you seen any items of particular interest?
- What is your primary objective in this area?
- Is there only one way to traverse?
- Have you tried moving the camera to get a better view?
- What does the X button do?
- What can do you to interact with that object?
- Have you completely searched this area?
    - Are you sure?

Problems:

1. This was hinted at earlier, but the participant wandered around, camped out in certain areas, and back-tracked through certain areas far more than I anticipated and I wasn't sure if I should probe more to get them on track and get more data or to stay quiet and let them wander but also limit potential data collection.
2. The participants, stopped verbalizing quite frequently and it got to the point where I felt like I may have been perceived as annoying by constantly reminding them to verbalize again.
3. There were a few cases where an objective was in the background but the participant didn't notice it, and rather than taking their inaction as a data point to address later, I may have prompted too much by helping them progress inorganically.

Potential Solutions:

1. The participant's gaming literacy is very low, so in the case that progress cannot be made without researcher intervention, perhaps a simpler alternative game can be prepared as a backup. Or, researcher intervention can be allowed with
2. I can look for cues of enjoyment versus frustration and proceed with probes accordingly. Thankfully, the participant in this instance was smiling the entire time, and they express their

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appreciation for the supportive prompts so I don't think I was annoying them, but I can see how this may be an issue for others.

   a. The way I was dynamically prompting the participant without ever giving them a direct answer and the participant's appreciation of this approach, which was far more responsive than the in-game support systems hinted at potential designs for more robust hint systems that might prove useful for less game-literate players.

3. It may come at the cost of useful data, and may lead to the player becoming more frustrated, but I can say less and let potential participant inability to progress and a strong signal that there is room for adjustment.

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Wrap-up

Lastly, after the think aloud session has concluded, the participant will be asked if they have anything left to add and subsequently thanked for their time.

DATA ANALYSIS PLAN

The results will be written up into a brief report that includes that participant's profile (results from learner profile questionnaire) and the play test data (utterances made at points of interest, utterances made between points of interest in each respective area). Lastly, results from the observation session, including pre-test, observation data and post-interview response highlights can all be cross-referenced, summarized and added to these think-aloud results upon request. If necessary, a brief slide show presentation can additionally be prepared and delivered upon request.

With more participants, the questionnaire and interview data can be collated and aggregated to try and identify user patterns.

FINAL REFLECTION

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After conducting the think-aloud focused research session, much potential for improvement was identified. I can redesign my observation form to better accommodate a wider variety and non-linear player behaviors, I can adjust the warm-up activity to facilitate more opportunities for verbalization and I can prepare a longer list of more organic prompts to encourage participants to speak more when they get focused on the game and stop speaking. Another minor issue was that when revisiting the audio recording it can be difficult to remember what exactly in the game was being referred to, but I suppose here is where video observation recordings (week 9's topic) will be of import.

Despite these issues, overall I think the session went well, and I even made some potentially interesting observations for future study. Though the warm-up can use adjustment it still did a good job of setting the expectation and also relaxing the participant before the main part of the session. The participant seemed to enjoy and respond better to plain language and specific prompts coming from the researcher compared the in-game prompts. And the participant had trouble seeing objects when they were located in the distance/ in the background indicating that those objects might benefit from being adjusted to stand out more (color, sound, animation, etc.), the camera/player position might benefit from adjustment, or the game itself might benefit from teaching the player via incentivizing/scaffolding/etc. toward searching the background for potential objects of interest.

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